

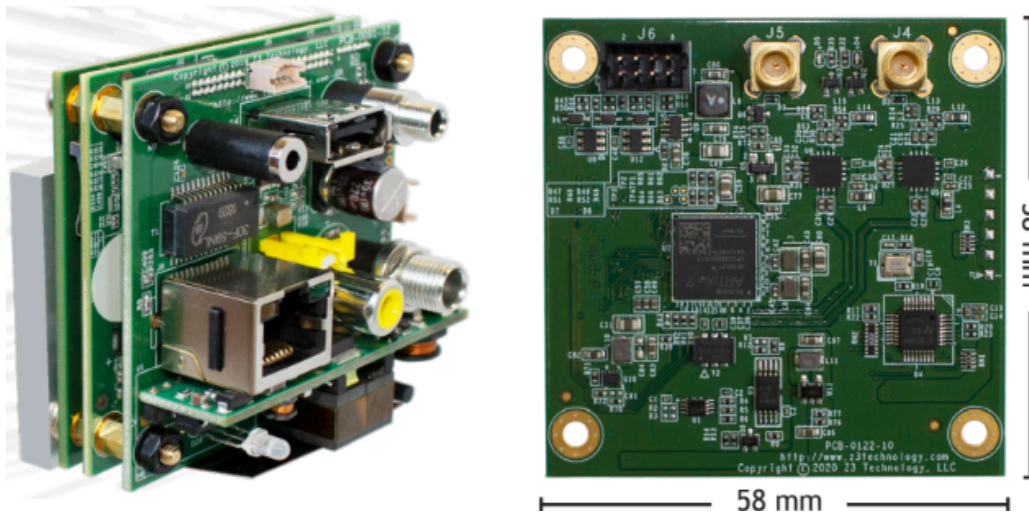
Dual Video Recorder (DVR)

The Dual Video Recorder (DVR) is designed for dual Composite / HDMI camera inspection, surveillance, industrial and UAV applications. The flexibility of the system allows for two video source or a visible camera and an infrared or night vision camera. The system is capable of encoding and streaming multiple channels of H.265 and H.264 video in HD resolutions. The Input interface board features flexible digital and analog input ports. The encoder can be configured to stream over IP and simultaneously record to USB or micro SD card.

Introduction

An indigenous Digital Video Recorder (DVR) is a specialized recording device designed to capture and store mission-critical video data in demanding defense environments. It is built with a rugged, military-grade enclosure capable of operating under extreme temperatures, vibration, dust, and electromagnetic interference typically encountered in field deployments.

This DVR plays an important role in air-defense operations by recording real-time video feeds from electro-optical sensors used for surveillance, target tracking, and engagement monitoring. It supports secure, time-stamped recording, reliable data storage, and quick retrieval for post-mission analysis and system diagnostics. Due to its reliability and seamless integration with defense control systems, it is widely used in modern indigenous air-defense platforms.



Architecture

The architecture of the indigenous Digital Video Recorder (DVR) is designed around a modular and high-reliability framework to ensure continuous recording of mission-critical video data. It typically consists of a video input interface that receives real-time feeds from electro-optical sensors, a high-performance processing unit for video encoding and compression, and a secure storage subsystem for long-duration recording. The system also includes time-synchronization modules, power conditioning circuits, and communication interfaces for integration with command and control networks. Built with ruggedized hardware and fault-tolerant design, the DVR architecture ensures stable operation under harsh field conditions while enabling secure data handling and quick retrieval for analysis, as per standards followed by organizations like the Defence Research and Development Organisation.

Why Choose Us?

 Rapid integration, faster time-to-market

 Clean, verified, documented RTL

 Direct access to experts



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Features

- **Professional Recording Capability:** Supports high-quality digital video recording, playback, and simultaneous recording with live viewing and system configuration.
- **Real-Time Video Capture:** Records high-resolution feeds from electro-optical surveillance and tracking sensors with minimal latency.
- **Rugged Military Design:** Built with a robust enclosure to operate under extreme temperatures, vibration, shock, dust, and electromagnetic interference.
- **Secure & High-Capacity Storage:** Provides reliable, tamper-resistant storage with support for long-duration recording and quick data retrieval.
- **Time Synchronization & Logging:** Enables precise time-stamped recording for accurate mission analysis and validation.
- **Advanced Processing & Integration:** Includes dedicated hardware for video encoding/compression and seamless interfacing with control and command systems.
- **Flexible Operation Features:** Supports trigger/pre-record modes, serial control option, external audio recording, and wide-range DC power operation (9–32V).
- **Salient Features:**
 - a) Low Latency, Low Power
 - b) Multiple serial connections available
 - c) Pass Camera Control Commands over IP
 - d) SRT Modes Supported: Caller, Sender, Rendezvous & SRT Encryption
 - e) Easy-to-Use HTTP-Based Control Interface
 - f) Simultaneous Streaming and Recording
 - g) Video Image Preview in the Web Gui
 - h) KLV Asynchronous and Synchronous Support

Specification

HARDWARE SPECIFICATIONS	
Parameter	Specification
Input Supply	• 28V DC
Communication	• RS-232, RS-422, RS-485, Ethernet 10/100/1G, USB
No. of Video Channels	• 02
Tracked Video IN	• PAL / CCIR standard (1V P-P on 75Ω Unbalanced)
Non-Tracked Video IN	
Recorded Video OUT	
Aspect Ratio	• 4:3
MPEG4 encoding Audio & Video	• H264



Specification Cont...

Video Quality	• Low, Mid, Hi
Video I/P	• Composite video X 2
Video Standard	• PAL/NTSC auto detection or digital HDMI/DVI
Audio I/P	• 2 x single ended mono/internal microphone
Video O/P	• Single analogue composite video
Audio O/P	• Audio Line Out
Control/ Remote	• Remote Control via serial port (option)
Download	• Via USB 2.0T through main connector or removable Flash Drive/ SSD card
Storage interface Internal or External	• 64 GB and above Upgradable to 500 GB
LCD Display	• Screen size 5.7 Inch
Elements	• Colour TFT LCD / LED Display
Resolution	• 640(W) x 480(H)
Power Requirement	• (9-32V DC) external power
Operation Temperature	• -25°C + 60°C
Mass	• 2.6 Kg (Approx.)
Environmental characteristics	• MIL-STD-810E
Electromagnetic Compliance	• MIL-STD-461D
Compatible with	• IP67

Get in touch with us

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